TECHNICAL PRESENTATION

ITA0508-Computer Vision

IMAGE MODELS

Impact of image formation models on application accuracy, defining related problems and using diagrams or simulations for explanation.

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IMAGE FORMATION MODELS

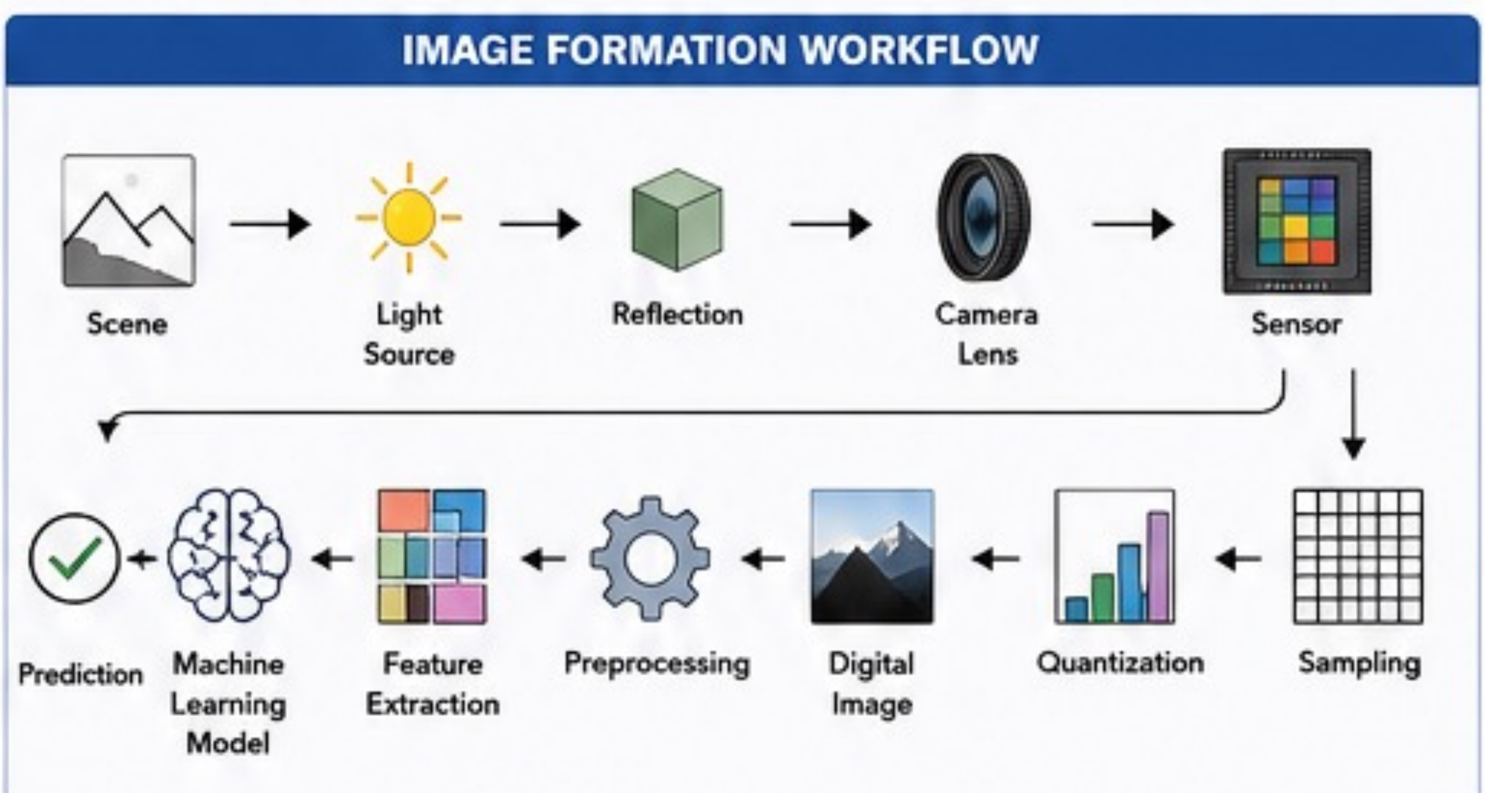
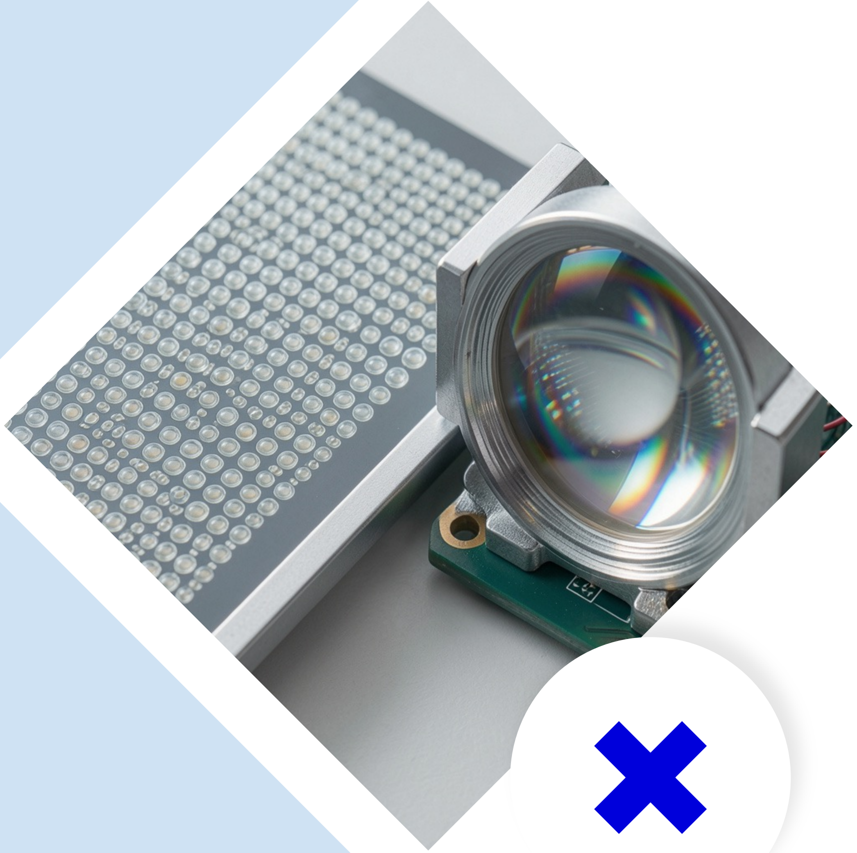


IMAGE FORMATION WORKFLOW
Scene
Light Source
Reflection
Camera Lens
Sensor
Sampling
Quantization
Digital Image
Preprocessing
Feature Extraction
Machine Learning Model
Prediction



PHYSICAL PRINCIPLES

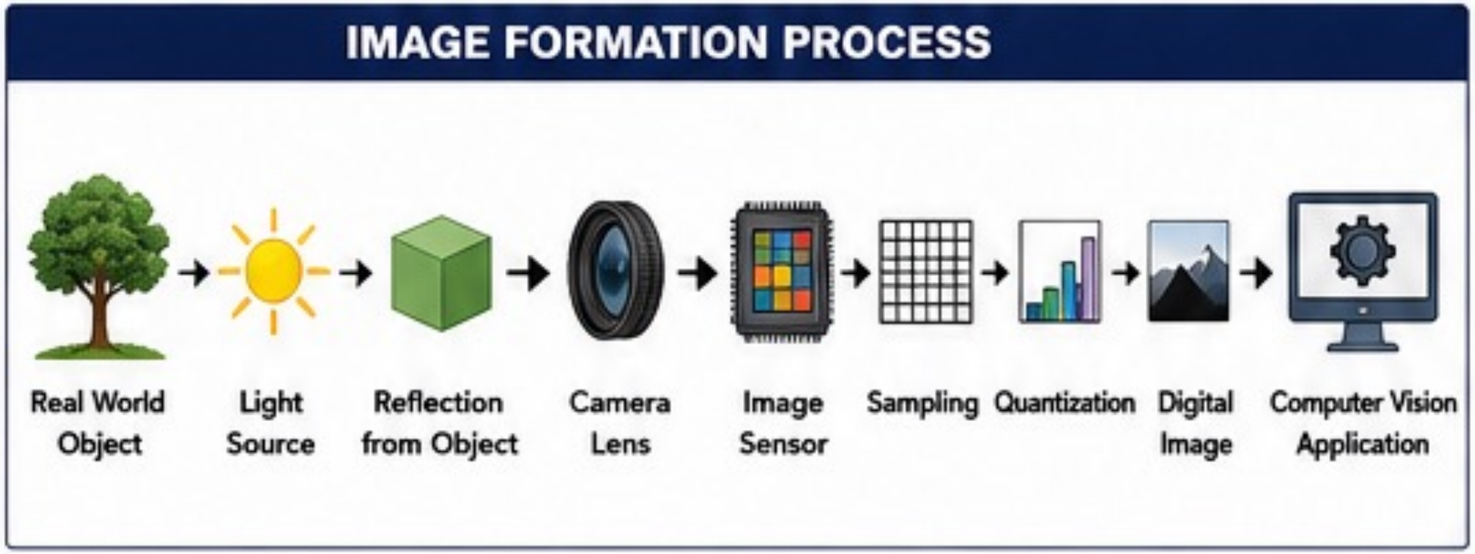
Radiometry, optics, illumination and sensor response govern image formation.

Point spread, geometry, and **Point Spread Function (PSF)** determine spatial blur, while sensor quantization and **noise** shape measurable fidelity.

MATHEMATICAL MODELS

Imaging often uses a *forward model* I = Hx + n capturing system blur H, scene x and noise n.

Analyze via **forward model** evaluation and **inverse problems** (reconstruction, parameter estimation). Use metrics like MSE and PSNR to quantify accuracy and guide regularization.



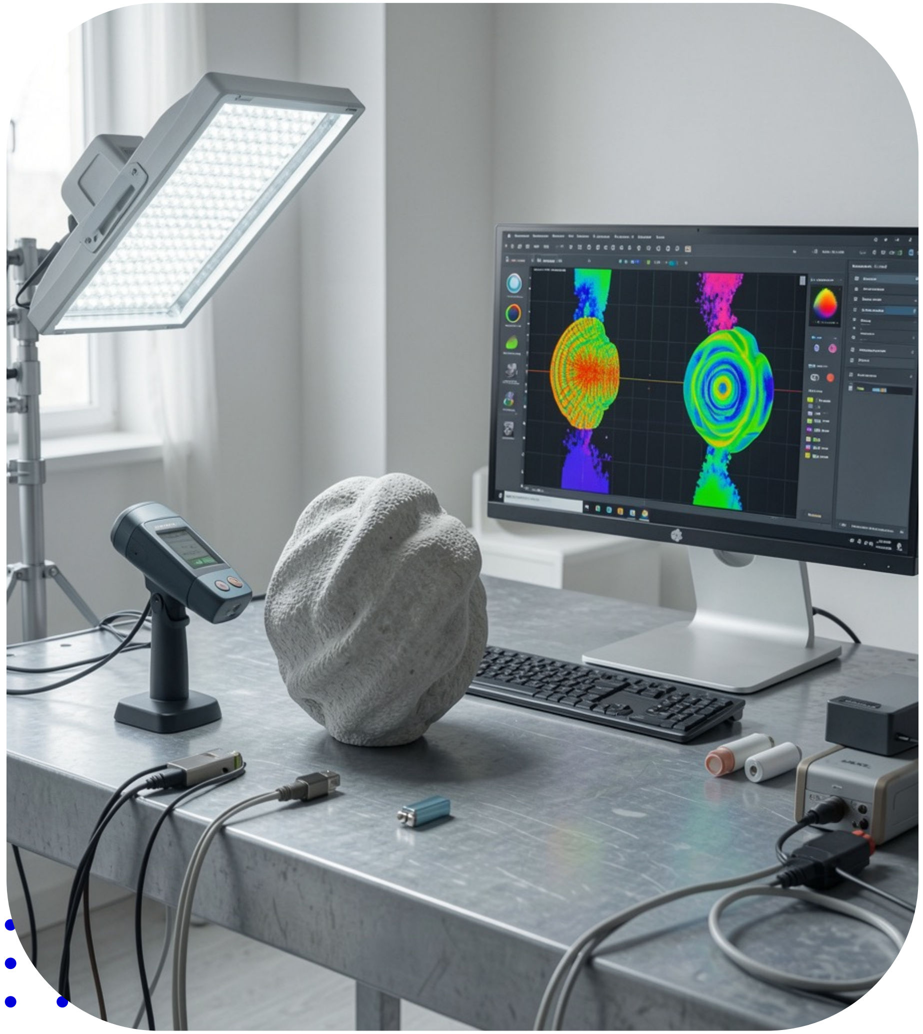
SIMULATION TOOLS

Use *ray tracing* and physics-based renderers to model light transport, optics, and sensor response.

Incorporate *noise models*, PSF simulation, and sensor quantization to produce realistic datasets for algorithm evaluation and validation.

IMPACT ON APPLICATION ACCURACY

Image Formation Factor	Effect on Image	Impact on Accuracy
Low Resolution	Loss of fine details	Lower detection accuracy
Blur	Edge information lost	Poor object recognition
Noise	Random pixel variations	False detections
Lens Distortion	Bent or warped image	Incorrect measurements
Poor Illumination	Images too dark or bright	Reduced classification accuracy
Motion Blur	Smearing of objects	Tracking failure
Incorrect Focus	Blurry objects	Low recognition rate



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APPLICATION ACCURACY IMPACT

ERROR SOURCES

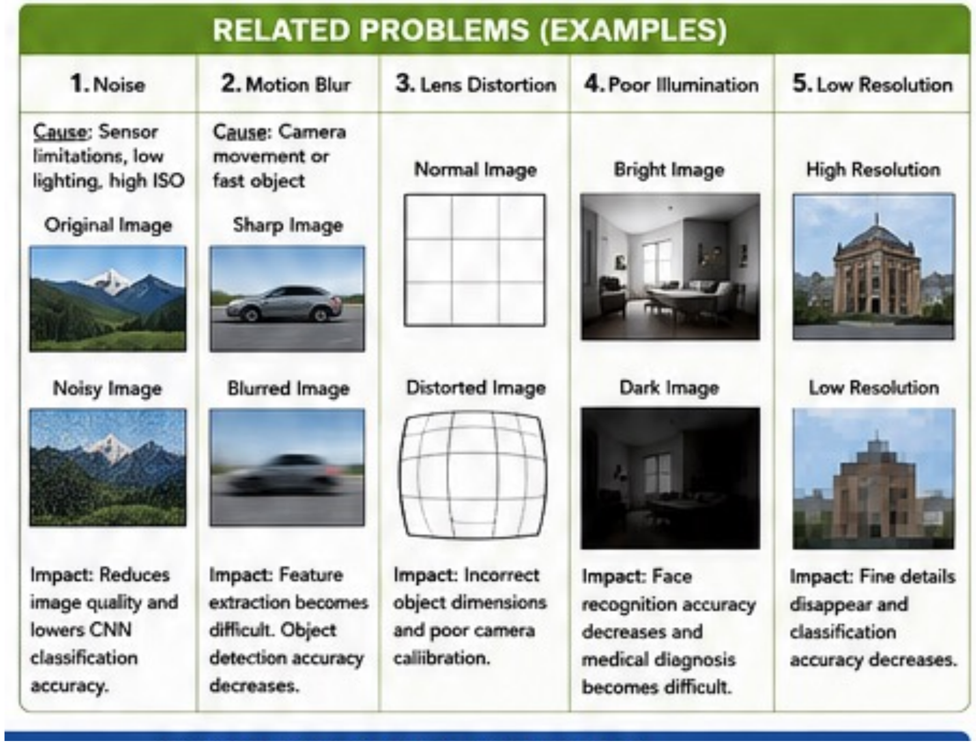
Model mismatch between assumed and real optics generates bias and artifacts.

Calibration errors, unmodeled illumination, sensor noise, and quantization reduce estimator reliability and degrade **task accuracy**.

PERFORMANCE METRICS

Evaluate with reconstruction metrics like **MSE** and **SSIM**, and task metrics such as classification or detection accuracy.

Use *uncertainty measures* and ROC/precision-recall curves for operational assessment.



Advantages of Accurate Image Formation Models

- Higher object detection accuracy
- Better image quality
- Improved medical diagnosis
- Reliable autonomous navigation
- Accurate 3D reconstruction
- Reduced image distortion
- Better feature extraction

CONCLUSIONS

Image formation models play a critical role in determining the accuracy of computer vision applications. Factors such as illumination, camera geometry, sampling, quantization, noise, blur, and lens distortion directly influence image quality and the reliability of subsequent image analysis. By using appropriate image formation models and applying preprocessing techniques such as denoising, deblurring, distortion correction, and image enhancement, the accuracy of applications like object detection, facial recognition, medical imaging, and autonomous driving can be significantly improved.

THANK YOU